



Partial least squares structural equation modeling using SmartPLS: a software review

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Abstract

In their effort to better understand consumer behavior, marketing researchers often analyze relationships between latent variables, measured by sets of observed variables. Partial least squares structural equation modeling (PLS-SEM) has become a popular tool for analyzing such relationships. Particularly the availability of SmartPLS, a comprehensive software program with an intuitive graphical user interface, helped popularize the method. We review the latest version of SmartPLS and discuss its various features. Our aim is to offer researchers with concrete guidance regarding their choice of a PLS-SEM software that fits their analytical needs.

Introduction

Structural equation models allow the specification of complex inter-relationships between observed and latent variables. Partial least squares, typically referred to a PLS structural equation modeling (PLS-SEM) or PLS path modeling (Tenenhaus et al. 2005; Hair et al. 2011), is one particularly prominent approach for estimating such models (Hwang et al. 2019; Khan et al. 2019), which has recently gained massive dissemination in business research and other scientific fields such as agriculture, ecology, environmental sciences, geography, and psychology (Sarstedt 2019). To illustrate this trend, we searched the Web of Science database for the emergence of the keywords “PLS-SEM” and “PLS path modeling” between 1995 and 2018. Figure 1 illustrates the resulting number of articles, which shows a pronounced upward trend that originated in the early 2000s, experiencing a massive surge in recent years.

While there is little doubt that many factors have contributed to this development—for instance, the publication of seminal articles on the method (Chin 1998; Tenenhaus et al. 2005; Hair et al. 2011)—the 2005 release of SmartPLS 2 (Ringle et al. 2005), a proprietary software package with graphical user interface to run PLS-SEM analyses, certainly has had a massive impact. SmartPLS 2 stands out in terms of ease-of-use, facilitates running standard PLS-SEM analyses, and is free of charge. Almost 15 years later, SmartPLS 3 (Ringle et al. 2015) has become a commercial software, offering a much broader range of modeling and algorithmic options, improved usability compared to version 2, and a professional support. In this article, we review the latest version (3.2.8) of SmartPLS.

A brief review of PLS-SEM

Model specification and estimation

Model specification in PLS-SEM involves the structural model and the measurement models. The structural model represents the structural paths between the constructs, whereas the measurement models represent the relationships between each construct and its associated indicators. Both models are typically visualized in the form of a path model, which graphically illustrates the hypotheses and variable relationships to be estimated in the PLS-SEM analysis.

The structural model is formally defined as:

$$\gamma = B\gamma + \zeta, \quad (1)$$

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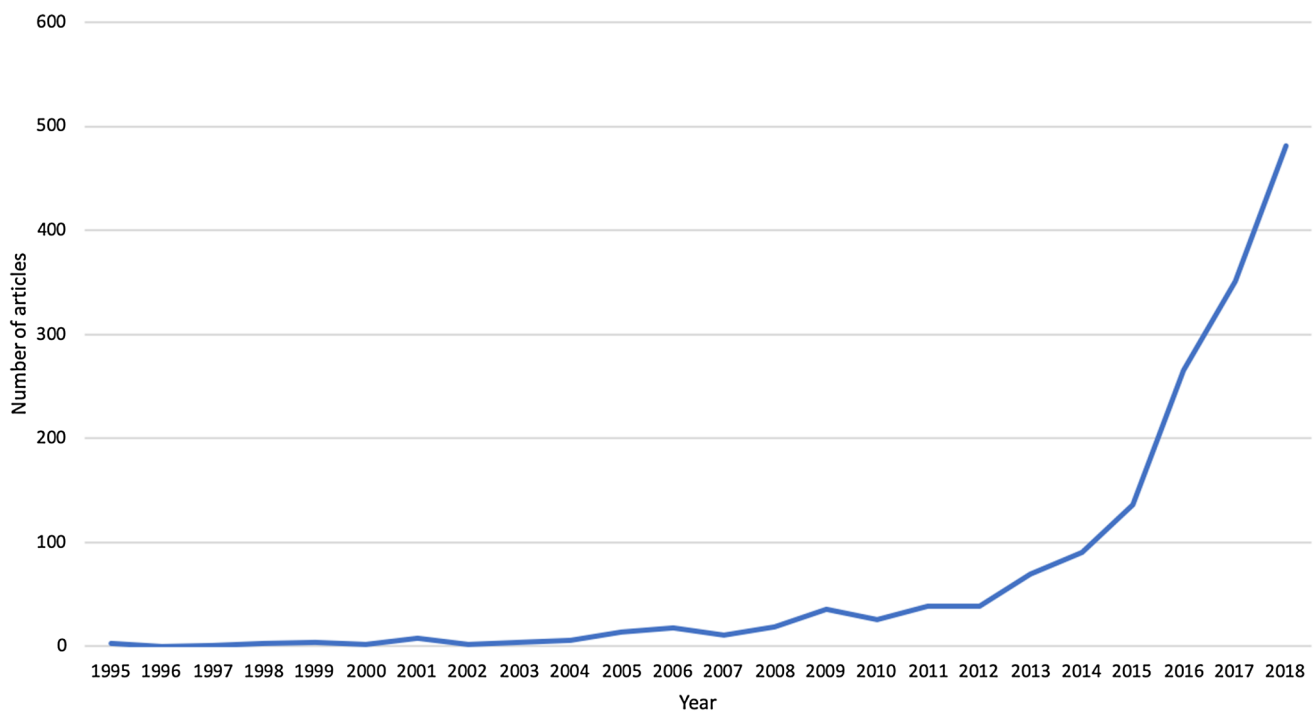


Fig. 1 Cumulative number of citations

where γ is a vector of all latent variables, $\mathbf{B}$ is a matrix of path coefficients, and ζ is the disturbance term of the dependent latent variables.

Measurement models can either be specified reflectively or formatively. In a reflective measurement model, the indicators are viewed as imperfect reflections of the underlying construct (MacKenzie et al. 2011). A reflective measurement model is given by

$$\mathbf{Z} = \mathbf{C}\gamma + \varepsilon, \quad (2)$$

where $\mathbf{z}$ is a vector of all indicators, $\mathbf{C}$ is a matrix of loadings relating indicators to latent variables, and ε is the disturbance term for indicators. When visualized in a path model, a reflective measurement model has direct relationships from the construct to its indicators.

In contrast, a formative measurement model combines indicators to form the construct (MacKenzie et al. 2011):

$$\gamma = \mathbf{H}\mathbf{z} + \theta, \quad (3)$$

where $\mathbf{H}$ is the matrix of weights from regressions of each latent variable on its associated indicators, and θ is the latent variables' disturbance term, which, however is set to zero, making the formative model equivalent to a weighted composite of indicators (Sarstedt et al. 2016). A formative measurement model has direct relationships from a set of indicators to a certain construct.

Model estimation in PLS-SEM—as implemented in SmartPLS—draws on Lohmöller's (1989) extension of

Wold's (1982) original PLS algorithm, which comprises two stages. The first stage applies a four-step algorithm to estimate the weights for each measurement model. The weights either correspond to zero-order correlations between a latent variable and each of its assigned indicators (Mode A) or result from regressing a latent variable on its associated indicators (Mode B). After convergence (i.e., between iterations, the sum of weights changes becomes very small; e.g., <0.0000001), the weights are used to compute the latent variable scores as linear combinations of their indicators.

The second stage uses the latent variable scores as input in a series of ordinary least squares regressions to estimate the measurement and structural model parameters (e.g., loadings, weights, and path coefficients). PLS-SEM therefore follows a composite-based approach to SEM in that the algorithm uses weighted composites of indicators to represent the latent variables in the statistical model (Rigdon et al. 2017)—regardless of whether the measurement models are specified reflectively or formatively (Sarstedt et al. 2016).

Various extensions of Lohmöller's (1989) PLS-SEM algorithm have been proposed to allow handling categorical variables (Jakobowicz and Derquenne 2007), testing for unidimensionality of measurement models (Menichelli et al. 2014), or adjusting the original estimates to accommodate common factor models (Dijkstra and Henseler 2015; Kock 2019). Other extensions have been proposed in the context of latent class analysis, a class of methods designed to identify and treat unobserved heterogeneity. The first approach



in this field is finite mixture PLS (FIMIX-PLS), which replaces the ordinary least squares regressions in the structural model with mixture regressions in the second stage of the PLS-SEM algorithm (Hahn et al. 2002). While FIMIX-PLS proves particularly useful for deciding on the number of segments to retain from the data (Sarstedt et al. 2011, 2019), more recently developed latent class methods such as PLS prediction-oriented segmentation (Becker et al. 2013) have been shown to perform superior in recovering the segment-specific model estimates (Schlittgen et al. 2016). Sarstedt et al. (2017b) have proposed using FIMIX-PLS and PLS-POS in tandem to identify and treat unobserved heterogeneity in PLS-SEM.

Model evaluation

Model evaluation in PLS-SEM follows a two-step process. The first step involves examining the measurement models and draws on different sets of metrics, depending on the measurement model type. In case of reflective measurement models, researchers need to assess the indicator and construct reliabilities, convergent validity, and discriminant validity (Sarstedt et al. 2017a; Hair et al. 2019a). Formative measurement models need to be assessed with regard to convergent validity, multicollinearity, and the significance and relevance of the indicator weights (Sarstedt et al. 2017a; Hair et al. 2019a). The second step involves examining the structural model and focuses on the significance and relevance of the path coefficients and the model's explanatory power (i.e., the R^2) as well as its predictive power (e.g., using PLSpredict; see Shmueli et al. 2016, 2019). Model fit testing using metrics such as SRMR or exact fit tests (Lohmöller 1989; Henseler et al. 2014) should be considered with extreme caution due to their conceptual shortcomings in the context of PLS-SEM (Hair et al. 2019a, b). Instead, researchers should focus on predictive model evaluation, which conforms to PLS-SEM's causal-predictive nature (Jöreskog and Wold 1982) and consider different model configurations by engaging in model comparisons (Sharma et al. 2019).

The SmartPLS software

Core features

After starting the software, users first need to initiate a project or to open an existing project file, which are clearly arranged in a separate window. Every project can accommodate multiple datasets and path models. This feature is particularly useful when users seek to estimate a model by means of different datasets such as in cross-cultural research, or seek to contrast different model variants using the same dataset. The software also comes with two sample model

files and datasets on corporate reputation (Schwaiger 2004) and the European customer satisfaction index (Eklöf and Westlund 2002), which allow users to readily start running the software. The corporate reputation model often serves as a showcase model for methodological advances (Hult et al. 2018; Sharma et al. 2019), many of which are implemented in the software, making it easy for novice users to use the corresponding techniques.

The software's graphical user interface is highly intuitive, comes in 15 different languages (e.g. Arabic, Malay, and Spanish) and enables novice users to quickly specify and estimate their path models. For instance, by dragging and dropping items from the indicators menu onto the modeling window, users can easily create new constructs with the selected items assigned. The constructs can then be linked by drawing single-headed arrows among them. The user interface also allows for including quadratic effects (Hair et al. 2018; Kumar and Purani 2018) and moderating effects without having to process the data outside SmartPLS, for example, to generate interaction terms (Becker et al. 2018). Figure 2 shows a screenshot of the graphical user interface, including the model estimates of the corporate reputation model.

Complementing Lohmöller's (1989) implementation of the PLS-SEM algorithm, including different structural model weighting schemes, and advanced settings to pre-specify initial item weights (see Hair et al. 2017b; Chapter 3), SmartPLS also allows running consistent PLS-SEM (PLS-SEM_c; Dijkstra and Henseler 2015), a variant of the original algorithm that accommodates factor models. Both versions of the PLS-SEM algorithm are supplemented by separate bootstrapping routines (Streukens and Leroi-Werelds 2016), which allow deriving standard errors for all model parameters. The bootstrapping routines are very comprehensive in that they offer users to request different confidence interval types, select among one- and two-tailed testing, and pre-specify significance levels. Missing value imputation procedures—albeit limited to mean replacement, casewise replacement, and pairwise replacement without considering more advanced options such as multiple imputation (Sarstedt and Mooi 2019)—and options to generate data groups complement the algorithmic implementations.

The standard results output, which can be saved in Excel and HTMT formats, is very well-arranged and allows users to easily browse between descriptive statistics (e.g., indicator and construct correlations) and state-of-the-art measurement as well as structural model evaluation metrics (e.g., Henseler et al. 2014; Franke and Sarstedt 2019; Sharma et al. 2019). Some of the metrics designed for testing a path model's predictive relevance—the blindfolding-based Q^2 and PLSpredict (Shmueli et al. 2016, 2019)—have to be requested using separate routines, which, however does not disrupt the frictionless exploration of new models. A particularly useful



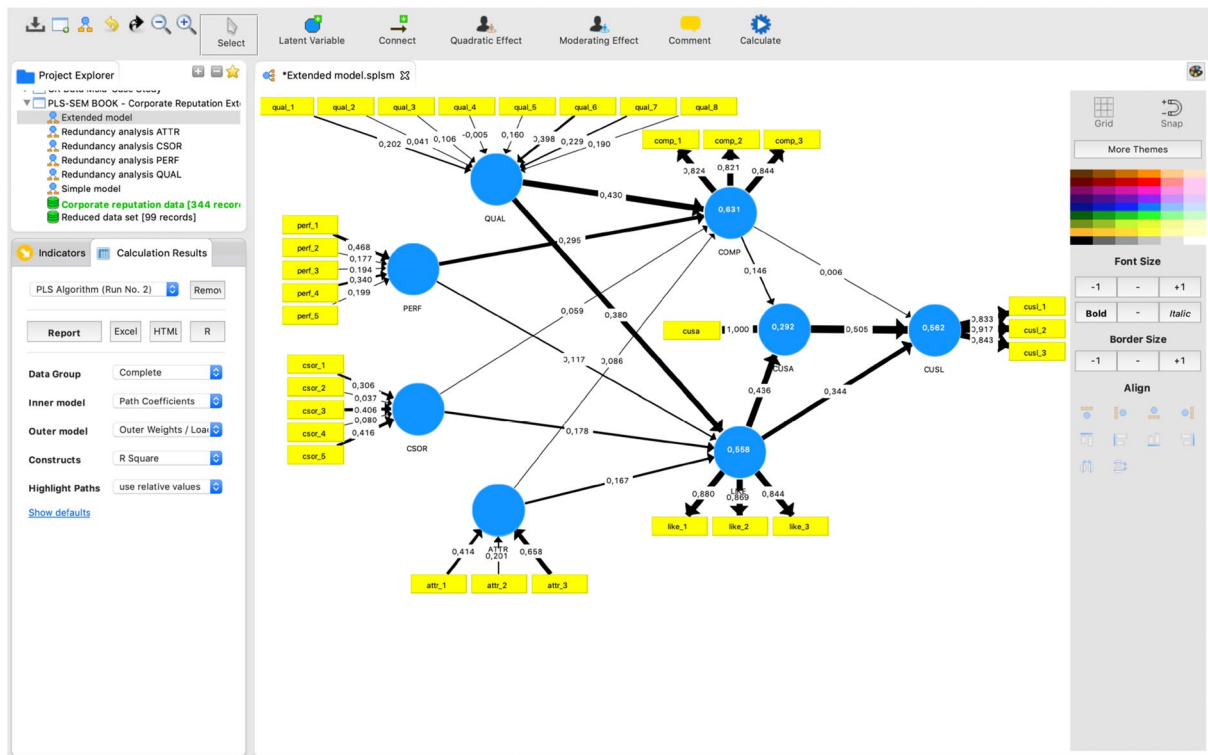


Fig. 2 SmartPLS graphical user interface with model estimates

feature is that most model evaluation metrics are colored in green, black, or red, depending on whether they meet generally recommended thresholds (Hair et al. 2017b; Sarstedt et al. 2017a). Thereby, the software lives up to PLS-SEM's original purpose to allow for "a dialogue between the investigator and the computer" (Wold 1985, p. 590), since tentative adjustments in the model set-up can quickly be evaluated in terms of their impact on the quality of measurement and structural models. This exploratory scope is also facilitated by several options implemented in the modeling window such as mouse-over information (e.g., R^2), highlighting of paths according to their relative size (Fig. 2), or navigating between different estimation runs.

The software also offers numerous extensions such as the confirmatory tetrad analysis, which allows for empirically testing the constructs' measurement mode (Gudergan et al. 2008), measurement invariance assessment (Henseler et al. 2016), approaches to multigroup analysis (Matthews 2017), and various methods for treating unobserved heterogeneity. A particularly beneficial feature in this regard is the option using FIMIX-PLS to identify an initial data partition (Sarstedt et al. 2011, 2019), which PLS-POS uses as input in a subsequent latent class analysis. Another important feature is the importance-performance map analysis (Ringle and Sarstedt 2016), which extends the standard PLS-SEM results reporting of path coefficient estimates

by adding a dimension to the analysis that considers the average values of the latent variable scores. The IPMA contrasts structural model total effects on a specific target construct (i.e., the importance, shown on the x-axis) with the average latent variable scores of this construct's predecessors (i.e., the performance, shown on the y-axis). Figure 3 illustrates an importance-performance map for the sample corporate reputation model included in the software.

Pricing

The most recent version of SmartPLS is commercial and can be purchased in the form of single user or network licenses. The developers offer a 30-day free trial version, which comes with all functionalities. The student version of SmartPLS is always available to every user but it has limited functionalities (e.g., no EXCEL and HTML results report exports) and is restricted to datasets with 100 cases. The professional version has all features and costs 200 € per year for academics. Users can also purchase shorter (1 month, 20 € for academics) or longer licenses (5 years, 920 € for academics). The developers also offer bulk discounts for up to 51 licenses for computer labs and commercial users.



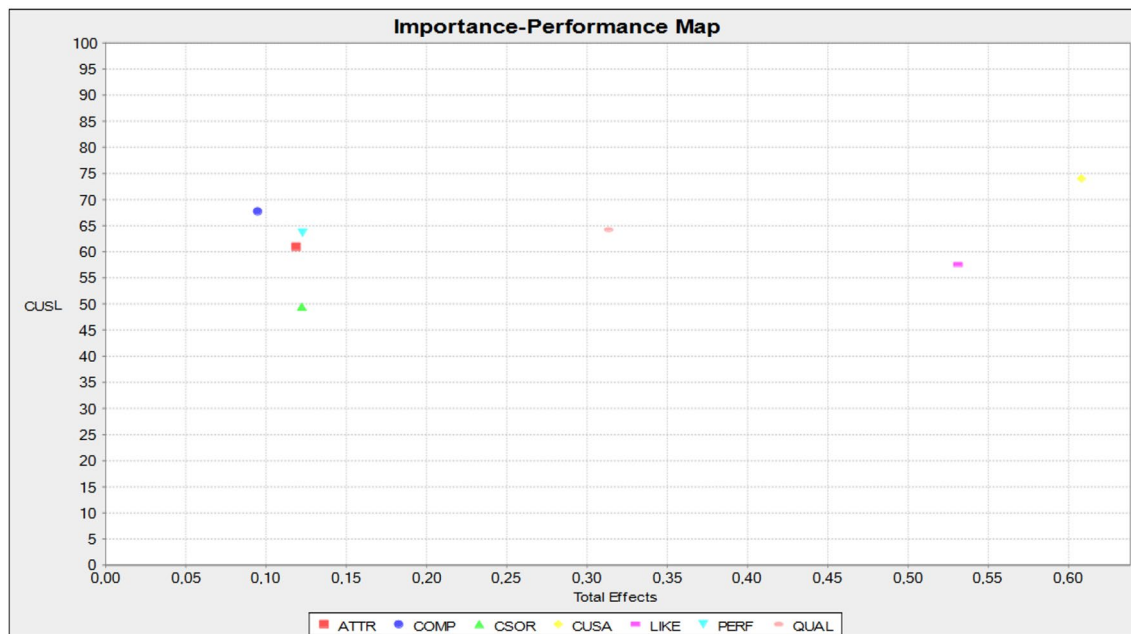


Fig. 3 Importance-performance map

Conclusion

SmartPLS is currently the most comprehensive software for conducting PLS-SEM analyses (Henseler 2017). In addition, SmartPLS is the most often used PLS-SEM software as several review studies of the method's application across disciplines reveal (Hair et al. 2017a; Usakli and Kucukergin 2018; Cepeda-Carrion et al. 2019). The software stands out due to its frictionless design, which also allows novice researchers to quickly specify and estimate PLS path models. Some researchers have lamented that “easy-to-use ‘front end’ software packages [...] made PLS usable by those with few statistical skills” and that “a simple-to-use automated tool would ‘crowd out’ more sophisticated techniques in fields such as information systems (Rouse 2009 pp. 2 and 4). Considering the strong emphasis that the information systems field pays to understanding the impact of users’ perceived ease-of-use on technology adoption (e.g., Venkatesh et al. 2003), the notion that a statistical program can only be considered rigorous when it is complicated to use seems rather bewildering. Rather, SmartPLS’s frictionless design follows best practice that is a condition sine qua non for modern software packages.

While several commercial alternatives exist (e.g., ADANCO, PLS-Graph, and WarpPLS), these are limited in scope, for example, with regard to language modules or algorithmic options, to some extent nontransparent with regard to the implementation of procedures, and are not regularly updated. Considering the rapid pace in methodological research on PLS-SEM (Hwang et al. 2019; Khan

et al. 2019), this latency can become problematic in journal review processes and may hinder publication success. On the contrary, the SmartPLS developers run regular updates, which not only include bug fixes, but extend the software’s functionalities. Of course, users can also revert to free of charge PLS-SEM modules in R or Stata (Monecke and Leisch 2012; Venturini and Mehmetoglu 2017), which offer much flexibility. These are, however, targeted at more experienced users with profound programming skills and come with highly technical documentation. On the contrary, numerous textbooks exist (Garson 2016; Ramayah et al. 2018; Wong 2019), which introduce PLS-SEM using screenshots from the SmartPLS software, thereby facilitating users’ adoption of even advanced topics (Hair et al. 2018). Hair et al.’s (2017b) *Primer on Partial Least Squares Structural Equation Modeling (PLS-SEM)*, which is strongly tied to the software and its language modules, has been translated into various languages including German, Spanish, and Vietnamese. Particularly novice users will benefit from such profound documentations of the software.

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